بسم الله الرحمن الرحيم

Discrete Mathematics

CHAPTER 2

First Order Logic

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Introduction to First Order Logic or Predicate Logic

From the previous lecture:

Everyone enrolled in the university has lived in a dormitory.

Mia has never lived in a dormitory.

$\therefore$ Mia is not enrolled in the university.

» Our target is to determine whether the above mentioned argument is correct or incorrect.

For this, we should understand Predicate Logic or First Order Logic.

In order to understand Predicate Logic, we must understand the following:

1. Predicates. ✓
2. Quantifiers. ✓

Predicates:

Definition: Predicates are the statements involving variables which are neither True nor False until or unless the values of the variables are specified.

For example:

x is an animal

x is greater than 3

x is less than 4

$x+y=7$

All the above statements are neither True nor False.

They are not propositions.

Consider the statement: "x is an animal"

Here x is a subject (something we are discussing about).

"is an animal" is a predicate (refers to a property that the subject of the statement can have).

In Predicate Logic, a statement is divided into two parts:

1. Subject ✓
2. Predicate ✓

Usually, we denote such statements with a shorthand notation.

For example: "x is greater than 3" can be represented by G(x)
where G() denotes the predicate "is greater than 3"
and x denotes the subject or variable.

After assigning the value of a variable x, the statement G(x) becomes a proposition and has a truth value (either True or False)

For example:

G(5) = 5 is greater than 3. (True)

G(3) = 3 is greater than 3. (False)

We will deal with more such examples in the next lecture. ☺

Quantifiers

Definition: Quantifiers are words that refer to quantities such as "some" or "all".
It tells for how many elements a given predicate is True.

In English, Quantifiers are used to express the quantities without giving an exact number.

Ex: all, some, many, none, few etc.

Sentences like - "Can I have some water?"
"Jack has many friends here."

DEFINITION 1 The *universal quantification* of $P(x)$ is the statement

"$P(x)$ for all values of x in the domain."

The notation $\forall x P(x)$ denotes the universal quantification of $P(x)$. Here $\forall$ is called the **universal quantifier.** We read $\forall x P(x)$ as "for all $x P(x)$" or "for every $x P(x)$." An element for which $P(x)$ is false is called a **counterexample** of $\forall x P(x)$.

TABLE 1 Quantifiers.

Statement	*When True?*	*When False?*
$\forall x P(x)$	$P(x)$ is true for every x.	There is an x for which $P(x)$ is false.
$\exists x P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x.

Types of Quantifiers:

1. Universal Quantifier.
2. Existential Quantifier.

Example: Let P(x) be a statement $x + 1 > x$

$P(1)$: $1+1 > 1 = 2 > 1$ (True)

$P(2)$: $2+1 > 2 = 3 > 2$ (True)

⋮

for all +ve integers x.

$P(x)$ is true for all +ve integers x. or $\forall x\, P(x)$ $\forall$

Quantifier.

for all.

Example 2: Let $Q(x)$ be the statement $x < 2$

$Q(1)$: $1 < 2$ (True)

$Q(2)$: $2 < 2$ (False)

$\vdots$

Q. Is there some value of x for which $Q(x)$ is True?

(If domain is set of +ve integers)

A. Yes. $Q(x)$ is true for $x = 1$.

Therefore, there is an x for which $Q(x)$ is True.

OR

$\exists x \, Q(x)$

There exists

There exists some x for which $Q(x)$ is true.

Universal Quantifier

Definition: The Universal Quantification of $P(x)$ is the statement

"$P(x)$ for all values of x in the domain"

Notation: $\forall x\ P(x)$

$\hookrightarrow$ for all x, $P(x)$

or

for Every x, $P(x)$

$\forall$ is called universal Quantifier

EXAMPLE 1 Let $P(x)$ denote the statement "$x > 3$." What are the truth values of $P(4)$ and $P(2)$?

Solution: We obtain the statement $P(4)$ by setting $x = 4$ in the statement "$x > 3$." Hence, $P(4)$, which is the statement "$4 > 3$," is true. However, $P(2)$, which is the statement "$2 > 3$," is false. ◀

EXAMPLE 3 Let $Q(x, y)$ denote the statement "$x = y + 3$." What are the truth values of the propositions $Q(1, 2)$ and $Q(3, 0)$?

Extra Examples

Solution: To obtain $Q(1, 2)$, set $x = 1$ and $y = 2$ in the statement $Q(x, y)$. Hence, $Q(1, 2)$ is the statement "$1 = 2 + 3$," which is false. The statement $Q(3, 0)$ is the proposition "$3 = 0 + 3$," which is true. ◀

EXAMPLE 9 Let $Q(x)$ be the statement "$x < 2$." What is the truth value of the quantification $\forall x Q(x)$, where the domain consists of all real numbers?

Solution: $Q(x)$ is not true for every real number x, because, for instance, $Q(3)$ is false. That is, $x = 3$ is a counterexample for the statement $\forall x Q(x)$. Thus

$$\forall x Q(x)$$

is false. ◀

EXAMPLE 12 What does the statement $\forall x N(x)$ mean if $N(x)$ is "Computer x is connected to the network" and the domain consists of all computers on campus?

Solution: The statement $\forall x N(x)$ means that for every computer x on campus, that computer x is connected to the network. This statement can be expressed in English as "Every computer on campus is connected to the network." ◀

Domain or Domain of Discourse:

A domain specifies the possible values of the variable under consideration.

For example: Let $P(x)$ is the statement "$x+1 > x$" and let us assume that

Domain -> set of all +ve integers.

$P(1) : 1+1 > 1$ True

$P(2) : 2+1 > 2$ True

Note: It is very important to specify the domain of discourse.
without it, universal quantification of a statement is not defined.

From the above example: $\forall x\, P(x)$ is true under the domain of +ve Integers.

Universal Quantifier - Counter Examples

Q. When does $\forall x\, P(x)$ becomes False?

A. $\forall x\, P(x)$ becomes False, when $P(x)$ is not always True
(where x is the domain under consideration)

Example 1: Let $P(x)$ be the statement "$x < 4$". What is the truth value of the quantification $\forall x\, P(x)$ where the domain consists of all +ve integers.

Solution: Finding just one counter example is enough to make $\forall x\, P(x)$ False.

$P(5)$: $5 < 4$ False.

As $P(5)$ is False. Therefore, $x = 5$ is one such counterexample for the statement $\forall x\, P(x)$. This shows that $P(x)$ is not True for every x in the domain of +ve integers.

Example 2: Let $Q(x)$ be the statement "$x + 1 > 2x$". If the domain consists of all integers, what is the truth value of $\forall x\, Q(x)$?

Solution: $Q(1)$: $1+1 > 2\times 1 \;=\; 2 > 2$ $\forall x\, Q(x)$ is false.

Homework Problem:

Q. If $R(x)$ is the statement "$x^2 \geq 0$"
$S(x)$ is the statement "$x^2 \geq x$"
and $T(x)$ is the statement "$x^2 < 0$"
Determine the truth values of $R(x)$, $S(x)$ and $T(x)$ if the domain of each variable consists of all integers.

Expressing Quantifications in English

Example: Let $P(x)$ be the statement "x spends more than five hours every weekday in class." where the domain for x consists of all students.
Express $\forall x\, P(x)$ in English.

Solution: $\forall x\, P(x)$:- for all x $P(x)$
for Every x $P(x)$
for Each x $P(x)$
for any x $P(x)$
all of x $P(x)$

1. For every x in the domain of all students, x spends more than five hours every weekday in class.
2. Every student spends more than five hours every weekday in class.

Universal Quantifier - Counter Examples

Q. When does $\forall x\, P(x)$ becomes False?

A. $\forall x\, P(x)$ becomes False, when $P(x)$ is not always True
(where x is the domain under consideration)

Example 1: Let $P(x)$ be the statement "$x < 4$". What is the truth value of the quantification $\forall x\, P(x)$ where the domain consists of all +ve integers.

Solution: Finding just one counter example is enough to make $\forall x\, P(x)$ False.

$P(5)$: $5 < 4$ False

As $P(5)$ is False. Therefore, $x = 5$ is one such counterexample for the statement $\forall x\, P(x)$. This shows that $P(x)$ is not True for every x in the domain of +ve integers.

Example 2: Let Q(x) be the statement "$x + 1 > 2x$". If the domain consists of all integers, what is the truth value of $\forall x\, Q(x)$?

Solution: $Q(1)$: $1 + 1 > 2 \times 1 = 2 > 2$ $\qquad \forall x\, Q(x)$ is false.

Homework Problem:

Q. If R(x) is the statement "$x^2 \geq 0$"
S(x) is the statement "$x^2 \geq x$"
and T(x) is the statement "$x^2 < 0$"
Determine the truth values of R(x), S(x) and T(x) if the domain of each variable consists of all integers.

Example 2: Let $N(x)$ be the statement "x has visited North Korea" where the domain consists of the students in your school. Express $\forall x\, N(x)$ in English.

Solution:

1. For every x in the domain of the students in your school, x has visited North Korea.
2. All students in your school has visited North Korea.

Homework Problem:

What does the statement $\forall x\, P(x)$ mean if $P(x)$ is "x is perfect" and the domain consists of people in your locality.

Existential Quantifier

Definition: The existential Quantification of $P(x)$ is the proposition "There exists an element x in the domain such that $P(x)$."

Or

$\exists x\, P(x)$

$\forall x\, P(x)$

Note:

Specifying the domain is important. Without domain, the statement $\exists x\, P(x)$ has no meaning.

Different ways to say $\exists x\, P(x)$:

"There is an x such that $P(x)$."

"There is at least one x such that $P(x)$."

"For some x $P(x)$."

Example 1: What is the truth value of $\exists x\, P(x)$, where $P(x)$ is the statement "$x^2 > 10$" and the universe of discourse consists of the positive integers not exceeding 4?

Solution:

$P(x)$: $x^2 > 10$ domain : $\{1, 2, 3, 4\}$ $\forall x\, P(x)$:

$P(1)$: $1^2 > 10$ False $P(4)$: $4^2 > 10$ True $P(1) \wedge P(2) \wedge P(3) \wedge P(4)$

$P(2)$: $2^2 > 10$ False

$\exists x\, P(x)$ is true.

$P(3)$: $3^2 > 10$ False

$\exists x\, P(x)$: $P(1) \vee P(2) \vee P(3) \vee P(4)$

Existential Quantifier - Examples

Example 2: Let $P(x)$ be the statement "$x = x^2$." If the domain consists of the integers. What is the truth value of $\exists x\, P(x)$?

Solution:

domain : Integers.

$P(x)$: $x = x^2$

$\exists x\, P(x)$? True

$P(1)$: $1 = 1^2$ True

Example 3: Let $Q(x)$ be the statement "$x + 1 > 2x$." If the domain consists of all integers, What is the truth value of $\exists x\, Q(x)$?

Solution:

$Q(x)$: $x + 1 > 2x$

domain : Integers.

$Q(-1)$: $-1 + 1 > -2 \Rightarrow 0 > -2$

$\exists x\, Q(x)$ is true

Example 4: Determine the truth value of each of these statements if the domain consists of all real numbers.

a) $\exists x\,(x^3 = -1)$ b) $\exists x\,(x^4 < x^2)$

Solution: a) $\exists x\,(x^3 = -1)$

let $x = -1$ $(-1)^3 = -1$ True

$\exists x\,(x^3 = -1)$ True.

b) $\exists x\,(x^4 < x^2)$

let $x = \frac{1}{2}$ $\left(\frac{1}{2}\right)^4 < \left(\frac{1}{2}\right)^2 \Rightarrow \frac{1}{16} < \frac{1}{4}$ True.

$\exists x\,(x^4 < x^2)$ is true.

Quantifiers with Restricted Domain

We can limit the domain of the quantifier by modifying the notation a bit.

For example: $\forall x < 0\ (x^2 > 0)$ Domain: Real Numbers

Meaning of the Above Statement: The Square of a negative real number is positive.

$\exists y > 0\ (y^2 = 2)$ Domain: Real Numbers

$\exists y > 0\ (y = \sqrt{2})$ $y = \sqrt{2}$ $\exists y > 0\ (y = \sqrt{2})$ is true.

Meaning of the above statement: There exists a positive square root of 2.

$$\forall x < 0\ (x^2 > 0) \Rightarrow \forall x\ (x < 0 \rightarrow x^2 > 0)$$

$$\exists y > 0\ (y^2 = 2) \Rightarrow \exists y\ (y > 0 \wedge y^2 = 2)$$

Logical Equivalences involving Predicates and Quantifiers

Definition: Two logical statements involving predicates and quantifiers are said to be equivalent if and only if they have the same truth value in all the possible cases.

Logical equivalence is helpful in replacing one expression with an equivalent expression.

There are two important equivalences involving quantifiers:

$$\forall x\, (P(x) \land Q(x)) \equiv \forall x\, P(x) \land \forall x Q(x)$$

$$\exists x\, (P(x) \lor Q(x)) \equiv \exists x\, P(x) \lor \exists x\, Q(x)$$

> Note: we can also use $\Leftrightarrow$ symbol instead of $\equiv$ symbol.

Why $\forall x\,(P(x) \wedge Q(x)) \equiv \forall x\,P(x) \wedge \forall x\,Q(x)$?

Proof: Let us consider one example

Domain: All students of xyz university.

$P(x)$: x has studied discrete mathematics.

$Q(x)$: x has scored more than 60% marks in exam.

$\forall x\,(P(x) \wedge Q(x))$: Every student in the xyz university has studied discrete mathematics and has scored more than 60% marks in exam.

$\forall x\,P(x) \wedge \forall x\,Q(x)$: Every student in the xyz university has studied discrete mathematics and every student in the xyz university has scored more than 60% marks in exam.

	Discrete Math	>60%
Mark	✓	✓
Joe	✓	✓

$\forall x(P(x) \wedge Q(x)) \rightarrow$

$\forall x\,P(x) \wedge \forall x\,Q(x)$

True.

What happens if we change $\wedge$ by $\vee$?

$\forall x\,(P(x) \vee Q(x)) \equiv \forall x\, P(x) \vee \forall x\, Q(x)$?

Proof: Domain: All students of xyz university.

$P(x)$: x has studied discrete mathematics.

$Q(x)$: x has scored more than 60% marks in exam.

$\forall x\,(P(x) \vee Q(x))$: Every student in the xyz university has studied discrete mathematics or has scored more than 60% marks in exam.

$\forall x\, P(x) \vee \forall x\, Q(x)$: Every student in the xyz university has studied discrete mathematics or every student in the xyz university has scored more than 60% marks in exam.

	Discrete Math	> 60%
Mark	✓	✗
Joe	✗	✓

Proof: Domain: All students of xyz university.

$P(x)$: x has studied discrete mathematics.

$Q(x)$: x has scored more than 60% marks in exam.

$\forall x\,(P(x) \vee Q(x))$: Every student in the xyz university has studied discrete mathematics or has scored more than 60% marks in exam.

$\forall x\, P(x) \vee \forall x\, Q(x)$: Every student in the xyz university has studied discrete mathematics or every student in the xyz university has scored more than 60% marks in exam.

	Discrete Math	> 60%
Mark	✓	✗
Joe	✗	✓

Conclusion: $\forall x\,(P(x) \wedge Q(x)) \equiv \forall x\, P(x) \wedge \forall x\, Q(x)$ True

$\forall x\,(P(x) \vee Q(x)) \equiv \forall x\, P(x) \vee \forall x\, Q(x)$ False

Logical Equivalences involving Predicates and Quantifiers

(Continued)

There are two important equivalences involving predicates:

$$\forall x\,(P(x) \wedge Q(x)) \equiv \forall x\, P(x) \wedge \forall x Q(x)$$

$$\exists x\,(P(x) \vee Q(x)) \equiv \exists x\, P(x) \vee \exists x\, Q(x)$$

Why $\exists x\,(P(x) \vee Q(x)) \equiv \exists x\, P(x) \vee \exists x\, Q(x)$?

Proof: Domain: All students of xyz university.

$P(x)$: x has studied discrete mathematics.

$Q(x)$: x has scored more than 60% marks in exam.

$\exists x\,(P(x) \wedge Q(x))$: Some students of xyz university has studied discrete mathematics and has scored more than 60% marks.

$\exists x\,P(x) \wedge \exists x\,Q(x)$: Some students of xyz university has studied discrete mathematics and some students of xyz university has scored more than 60% marks.

	Discrete Math	> 60%
Mark	✓	✓
Joe	×	×

$\exists x\,(P(x) \wedge Q(x))$: Some students of xyz university has studied discrete mathematics and has scored more than 60% marks.

$\exists x\, P(x) \wedge \exists x\, Q(x)$: Some students of xyz university has studied discrete mathematics and some students of xyz university has scored more than 60% marks.

	Discrete Math	> 60%
Mark	✓	✗
Joe	✗	✓

Conclusion: $\exists x\,(P(x) \wedge Q(x)) \equiv \exists x\, P(x) \wedge \exists x\, Q(x)$ False
$\exists x\,(P(x) \vee Q(x)) \equiv \exists x\, P(x) \vee \exists x\, Q(x)$ True

Negating Quantified Expression (Part 1)

Consider the following statement:

"Every student in the xyz university has studied discrete mathematics."

Domain: All students of xyz university.

P(x): x has studied discrete mathematics.

"Every student in the xyz university has studied discrete mathematics."

is equivalent to

$\forall x\, P(x)$

What is the negation of the above statement?

"It is not the case that every student in the xyz university has studied discrete mathematics."

OR

"There is some student in the xyz university who has not studied discrete mathematics."

P(x): x has studied discrete mathematics.

¬ P(x): x has not studied discrete mathematics.

"There is some student in the xyz university who has not studied discrete mathematics."

is equivalent to

$\exists x \neg P(x)$

So, the negation of $\forall x\, P(x)$ is $\neg\forall x\, P(x)$ or $\exists x \neg P(x)$.

1. Negating a Universal Quantifier ($\forall$):

The negation of "for all x, $P(x)$" is logically equivalent to "there exists an x such that $P(x)$ is false."

$$\neg(\forall x\, P(x)) \equiv \exists x\, \neg P(x)$$

Example:

- **Original**: "All students passed the exam." ($\forall x\, P(x)$)
- **Negation**: "There exists a student who did not pass the exam." ($\exists x\, \neg P(x)$)

1. Negating a Universal Quantifier ($\forall$):

The negation of "for all x, $P(x)$" is logically equivalent to "there exists an x such that $P(x)$ is false."

$$\neg(\forall x\, P(x)) \equiv \exists x\, \neg P(x)$$

Example:

- **Original**: "All students passed the exam." ($\forall x\, P(x)$)
- **Negation**: "There exists a student who did not pass the exam." ($\exists x\, \neg P(x)$)

2. Negating an Existential Quantifier ($\exists$):

The negation of "there exists an x such that $P(x)$" is logically equivalent to "for all x, $P(x)$ is false."

$$\neg(\exists x\, P(x)) \equiv \forall x\, \neg P(x)$$

Example:

- **Original**: "There exists a student who passed the exam." ($\exists x\, P(x)$)
- **Negation**: "All students did not pass the exam." ($\forall x\, \neg P(x)$)

Example 1:

- **Statement:** "All cars are electric." ($\forall x\, C(x)$)
- **Negation:** "There exists a car that is not electric." ($\exists x\, \neg C(x)$)

Example 2:

- **Statement:** "There exists a person who is happy." ($\exists x\, H(x)$)
- **Negation:** "All people are not happy." ($\forall x\, \neg H(x)$)

Example 3:

- **Statement:** "For all numbers x, $x^2 \geq 0$." ($\forall x\, x^2 \geq 0$)
- **Negation:** "There exists a number x such that $x^2 < 0$." ($\exists x\, x^2 < 0$)

Negating Quantified Expressions (Part 2)

Negating an existential quantification:

Domain: All students of xyz university.

$P(x)$: x has studied discrete mathematics.

"There is a student in xyz university who has studied discrete mathematics."

is equivalent to

$$\exists x\, P(x)$$

The negation of the above statement:

"It is not the case that there is a student in the xyz university who has studied discrete mathematics."

is equivalent to

$$\neg \exists x\, P(x)$$

OR

"Every student in the xyz university has not studied discrete mathematics."

"Every student in the xyz university has not studied discrete mathematics."

is equivalent to

$$\forall x\, \neg P(x)$$

Conclusion:

$$\left.\begin{aligned} \neg \forall x\, P(x) &\equiv \exists x\, \neg P(x) \\ \neg \exists x\, P(x) &\equiv \forall x\, \neg P(x) \end{aligned}\right\} \text{De Morgan's laws for quantifiers}$$

Negating Quantified Expressions

(Examples)

Example 1: What is the negation of the statement "All dogs bark"?

Solution:

$$
\begin{tikzcd}
\text{All dogs bark} \arrow[d] & \text{Some dogs do not bark} \\
\forall \text{ dogs } d,\ d \text{ barks} \arrow[d] & \exists \text{ dogs } d,\ d \text{ does not bark} \arrow[u] \\
\neg\,(\forall \text{ dogs } d,\ d \text{ barks}) \arrow[r, Rightarrow] & \exists \text{ dogs } d,\ \neg(d \text{ barks}) \arrow[u]
\end{tikzcd}
$$

Example 2: What is the negation of the statement "Most cars are inexpensive"?

Solution:

Most cars are inexpensive → ∃ cars c, c is inexpensive → ¬ (∃ cars c, c is inexpensive) → ∀ cars c, ¬(c is inexpensive) → ∀ cars c, c is not inexpensive → Not all cars are inexpensive

Homework problem:

Express the negation (in English) of each of the following statements:

1) All dogs have fleas.
2) There is a horse that can add.
3) No rabbits knows calculus.

Translating English Sentences to Logical Expressions

Let us suppose we have to translate the following English sentence into an equivalent logical expression.

"Every student of XYZ University has studied discrete mathematics."

First, we will rewrite the statement so that we can identify the appropriate quantifiers to use.

"For every student x in the XYZ University, x has studied discrete mathematics."

Let us suppose that the domain consists of all students of XYZ University.
And, S(x) denotes "x has studied discrete mathematics."

$$\forall x\ S(x)$$

Note:

We can also use a different domain of discourse and a set of predicates to form the equivalent logical expression of the above statement.

Domain: All people on the planet earth.

We need to express our statement as:

> "For every person x, if person x is a student in the XYZ University then x has studied discrete mathematics."

"For every person x, if person x is a student in the XYZ University then x has studied discrete mathematics."

Let us suppose that

P(x) denotes "person x is a student in the XYZ University."
S(x) denotes "person x has studied discrete mathematics."

$$\boxed{\forall x\, (P(x) \rightarrow S(x))}$$

Note: We can't write $\forall x\, (P(x) \land S(x))$ because this statement says that "All people are students in the XYZ University and has studied discrete mathematics."

We can also use a two-variable predicate S(x, y) which represents the statement "person x has studied subject y."

$$\forall x\, (P(x) \rightarrow S(x, \text{discrete mathematics}))$$

where the domain consists of all people on the planet earth.

$$\forall x\, (S(x, \text{discrete mathematics}))$$

where the domain consists of all students of XYZ University.

Translating English Sentences to Logical Expressions

(Part 2)

Let us suppose that we want to translate the following into logical expression:

"Some students in the XYZ University has studied discrete mathematics."

Let us consider that the domain consists of all students of XYZ University.

Domain: All students of XYZ University.
P(x): x has studied discrete mathematics.

$$\exists x\, P(x)$$

Let's change the domain.

Domain: All people on the planet earth.

$S(x)$: x is a student of XYZ University.

$P(x)$: x has studied discrete mathematics.

$$\exists x\,(S(x) \wedge P(x))$$

"There is some person x having the properties that x is the student of XYZ University and x has studied discrete mathematics."

Homework Problem:

Express the following statement using predicates and quantifiers:
"There is a student who has taken more than 21 credit hours in a semester and received all A's."

Translating English Sentences to Logical Expressions

(Part 3)

Problem: Consider these statements. The first two are called premises and the third is called the conclusion. The entire set is called an argument.

"All lions are fierce."
"Some lions do not drink coffee."
"Some fierce creatures do not drink coffee."

Let P(x), Q(x) and R(x) be the statements "x is a lion," "x is a fierce," and "x drinks coffee" respectively. Assuming that the domain consists of all creatures, express the statements in the argument using predicates P(x), Q(x), and R(x).

Solution: **Domain:** All creatures

P(x): x is a lion
Q(x): x is a fierce
R(x): x drinks coffee

$\forall x\,(P(x) \rightarrow Q(x))$

$\exists x\,(P(x) \land \neg R(x))$

$\exists x\,(Q(x) \land \neg R(x))$

Translating English Sentences to Logical Expressions

(Part 4)

Problem: Consider these statements, of which the first three are premises and the fourth is a valid conclusion.

"All hummingbirds are richly colored."
"No large birds live on honey."
"Birds that do not live on honey are dull in color."
"Hummingbirds are small."

Let $P(x)$, $Q(x)$, $R(x)$ and $S(x)$ be the statements "x is a humming bird," "x is large," "x lives on honey," and "x is richly colored" respectively. Assuming that the domain consists of all birds, express the statements in the argument using quantifiers and $P(x)$, $Q(x)$, $R(x)$, and $S(x)$.

Solution: **Domain:** All birds

$P(x)$: x is a humming bird
$Q(x)$: x is large
$R(x)$: x lives on honey
$S(x)$: x is richly colored

$$\forall x\,(P(x) \rightarrow S(x))$$

$$\begin{aligned}\forall x\,(Q(x) \rightarrow \neg R(x)) &\equiv \forall x\,(\neg Q(x) \vee \neg R(x)) \\ &\equiv \forall x\,\neg(Q(x) \wedge R(x)) \\ &\equiv \neg\exists x\,(Q(x) \wedge R(x))\end{aligned}$$

$$\forall x\,(\neg R(x) \rightarrow \neg S(x))$$

$$\forall x\,(P(x) \rightarrow \neg Q(x))$$

First Order Logic (Solved Problem 1)

Q. What is the logical translation of the following statement?

"None of my friends are perfect."

(A) $\exists x(F(x) \wedge \neg P(x))$

(B) $\exists x(\neg F(x) \wedge P(x))$

(C) $\exists x(\neg F(x) \wedge \neg P(x))$

(D) $\neg \exists x(F(x) \wedge P(x))$

[GATE CS 2013]

Solution: Domain: All people in this world.

Let $F(x)$ denotes "x is my friend."
and $P(x)$ denotes "x is perfect."

"None of my friends are perfect." can be translated to

$$\forall x(F(x) \rightarrow \neg P(x))$$
$$\equiv \forall x(\neg F(x) \vee \neg P(x))$$
$$\equiv \forall x \neg(F(x) \wedge P(x))$$
$$\equiv \neg \exists x(F(x) \wedge P(x))$$

$\therefore$ option (D) is the correct option.

First Order Logic (Solved Problem 2)

Q. Which one of the following is NOT logically equivalent to $\neg\exists x(\forall y(\alpha) \land \forall z(\beta))$?

(A) $\forall x(\exists z(\neg\beta) \rightarrow \forall y(\alpha))$ (B) $\forall x(\forall z(\beta) \rightarrow \exists y(\neg\alpha))$

(C) $\forall x(\forall y(\alpha) \rightarrow \exists z(\neg\beta))$ (D) $\forall x(\exists y(\neg\alpha) \rightarrow \exists z(\neg\beta))$ [GATE CS 2013]

Solution: $\neg\exists x(\forall y(\alpha) \land \forall z(\beta))$ can be rewritten as:

$$\forall x \neg(\forall y(\alpha) \land \forall z(\beta))$$
$$\equiv \forall x(\neg\forall y(\alpha) \lor \neg\forall z(\beta))$$
$$\equiv \forall x(\exists y(\neg\alpha) \lor \exists z(\neg\beta))$$

(A) $\forall x(\exists z(\neg\beta) \rightarrow \forall y(\alpha)) \equiv \forall x(\neg\exists z(\neg\beta) \lor \forall y(\alpha)) \equiv \forall x(\forall z(\beta) \lor \forall y(\alpha))$

(B) $\forall x(\forall z(\beta) \rightarrow \exists y(\neg\alpha)) \equiv \forall x(\neg\forall z(\beta) \lor \exists y(\neg\alpha)) \equiv \forall x(\exists z(\neg\beta) \lor \exists y(\neg\alpha))$

(C) $\forall x(\forall y(\alpha) \rightarrow \exists z(\neg\beta)) \equiv \forall x(\neg\forall y(\alpha) \lor \exists z(\neg\beta)) \equiv \forall x(\exists y(\neg\alpha) \lor \exists z(\neg\beta))$

(D) $\forall x(\exists y(\neg\alpha) \rightarrow \exists z(\neg\beta)) \equiv \forall x(\neg\exists y(\neg\alpha) \lor \exists z(\neg\beta)) \equiv \forall x(\forall y(\alpha) \lor \exists z(\neg\beta))$

$\therefore$ both (A) and (D) are correct options.

First Order Logic (Solved Problem 3)

Q. What is the correct translation of the following statement into mathematical logic?

"Some real numbers are rational"

(A) $\exists x(\text{real}(x) \vee \text{rational}(x))$

(B) $\forall x(\text{real}(x) \rightarrow \text{rational}(x))$

(C) $\exists x(\text{real}(x) \wedge \text{rational}(x))$

(D) $\exists x(\text{rational}(x) \rightarrow \text{real}(x))$

[GATE CS 2012]

Solution: Domain: All numbers

(A) There exist some numbers which are either real or rational.

(B) All real numbers are rational.

(C) There exist some numbers which are both real and rational.

(D) There exist some numbers such that if they are rational then they are real.

$\therefore$ option (C) is the correct option.

First Order Logic (Solved Problem 4)

Q. Which one of the following is the most appropriate logical formula to represent the statement?

"Gold and silver ornaments are precious".

The following notations are used:
$G(x)$: x is a gold ornament
$S(x)$: x is a silver ornament
$P(x)$: x is precious

(A) $\forall x(P(x) \rightarrow (G(x) \wedge S(x)))$
(B) $\forall x((G(x) \wedge S(x)) \rightarrow P(x))$
(C) $\exists x((G(x) \wedge S(x)) \rightarrow P(x))$
(D) $\forall x((G(x) \vee S(x)) \rightarrow P(x))$

[GATE CS 2009]

Solution: Domain: All ornaments.

(A) For all ornaments x, if x is precious then it is both gold and silver.
(B) For all ornaments x, if x is both gold and silver then it is precious.
(C) For some ornaments x, if x is both gold and silver then it is precious.
(D) For all ornaments x, if x is either gold or silver then it is precious.

$\therefore$ option (D) is the correct option.

Introduction to Nested Quantifiers

Defintion: Two quantifiers are said to be nested if one is within the scope of the other.

For example: $\forall x \exists y\, Q(x, y)$

$\exists$ is within the scope of $\forall$

Note: Anything within a scope of the quantifier can be thought of as a propositional function.

$$\forall x \underbrace{\boxed{\exists y\, Q(x, y)}}_{P(x)} \Rightarrow \forall x\, P(x)$$

Different combinations of Nested Quantifiers

Order of quantifiers doesn't matter $\leftarrow$

- $\forall x \forall y\, Q(x, y)$
- $\forall x \exists y\, Q(x, y)$
- $\exists y \forall x\, Q(x, y)$
- $\exists x \exists y\, Q(x, y)$

$\rightarrow$ Order of quantifiers does matter

i.e. $\forall x \forall y\, Q(x, y) \equiv \forall y \forall x\, Q(x, y)$
and $\exists x \exists y\, Q(x, y) \equiv \exists y \exists x\, Q(x, y)$

Also, $\forall x \exists y\, Q(x, y) \not\equiv \exists y \forall x\, Q(x, y)$

Nested Quantifiers (Example 1)

Example 1: Let x and y be the real numbers and $P(x, y)$ denotes "$x + y = 0$."

Find the truth values of

a) $\forall x \forall y\, P(x, y)$
b) $\forall x \exists y\, P(x, y)$
c) $\exists y \forall x\, P(x, y)$
d) $\exists x \exists y\, P(x, y)$

Solution: Domain: All real numbers.

a) $\forall x \forall y\, P(x, y) \equiv \forall x \forall y\, (x + y = 0)$

"For all real numbers x and y, $x+y = 0$"

Think of it as "For any combination of x and y, $P(x, y)$."

Is it True?

No.

Example: $1 + 2 \neq 0$

$\forall y \forall x\, P(x, y) \equiv \forall y \forall x\, (x + y = 0)$

"For all real numbers y and x, $x + y = 0$" ← Similar to the previous statement

b) $\forall x \exists y\, P(x, y) \equiv \forall x \exists y\, (x + y = 0)$

"For every real number x, there exist a real number y such that $x + y = 0$"

Is it True?

Yes.

You can choose any real number x; there is always a real number y for that x such that $x + y = 0$.

For example: For

$x = 1$	$x = -1$	$x = \frac{1}{2}$
$y = -1$	$y = 1$	$y = -\frac{1}{2}$

Steps:

1. Consider all different values of x.
2. Find just 1 value of y for each x such that $P(x, y)$ becomes True.

Note: Value of y depends on the value of x.

c) $\exists y \forall x\, P(x, y) \equiv \exists y \forall x\, (x + y = 0)$

"There exist some real number y such that for every real number x, $x + y = 0$"

Is it True?

No.

It is asking us to find a real number y for which $P(x, y)$ becomes true by plugging in every real number x.

What is $P(x, y)$ in our example?

$$P(x, y) \Rightarrow x + y = 0$$

First, take some real number y

Lets take $y = 1$

$$P(x, 1) \Rightarrow x + 1 = 0$$

Now, plug in all real numbers x.

$P(\frac{1}{2}, 1)$ is False, $P(1, 1)$ is False and so on.

No matter what y you choose, $P(x, y)$ is always **False** for all real numbers x.

d) $\exists x \exists y\, P(x, y) \equiv \exists x \exists y\, (x + y = 0)$

"There exist some real numbers x and y such that $x + y = 0$"

Obviously. There exist some combination of real numbers x and y exist for which $x + y = 0$.

Take $x = 1$ and $y = -1$ (Many such combinations)

$$\boxed{1 + (-1) = 0 \text{ is True.}}$$

$\therefore \exists x \exists y\, P(x, y)$ is True.

Also, $\exists x \exists y\, P(x, y) \equiv \exists y \exists x P(x, y)$

Nested Quantifiers (Example 2)

Example 2: Let x and y be the real numbers and $Q(x, y)$ denotes "$x.y = 0$."
Find the truth values of the following:

a) $\forall x \forall y\, Q(x, y)$
b) $\forall x \exists y\, Q(x, y)$
c) $\exists y \forall x\, Q(x, y)$
d) $\exists x \exists y\, Q(x, y)$

Solution: a) $\forall x \forall y\, Q(x, y) \equiv \forall x \forall y (x.y = 0)$

"For all real numbers x and y, $x.y = 0$"

Is it True? No.

For example: take $x = 1$ and $y = 2$

$$1.2 \neq 0$$

$x.y = 0$ is not satisfied for every combination of real numbers x and y.

b) $\forall x \exists y\, Q(x, y) \equiv \forall x \exists y\, (x.y = 0)$

"For every real number x there exist a real number y such that $x.y = 0$"

Is it True? Yes.

Example:

$x = 1$, $y = 0$	$x = -1$, $y = 0$	$x = \frac{1}{2}$, $y = 0$
$\downarrow$	$\downarrow$	$\downarrow$
$1.0 = 0$	$-1.0 = 0$	$\frac{1}{2}.0 = 0$

No matter what value of x is chosen, one can always take $y = 0$ to make $x.y = 0$.

c) $\exists y \forall x\, Q(x, y) \equiv \exists y \forall x\, (x.y = 0)$

"There exist some real number y such that for every real number x, $x.y = 0$"

Is it True? **Yes.**

I prefer to convert the above sentence in this form:

Find a real number y such that $Q(x, y)$ becomes true by plugging in all real numbers x.

Step 1: Find a real number y.

Let say $y = 0$.

Step 2: Put y in $Q(x, y)$

$$Q(x, y) = Q(x, 0) \Rightarrow x.0 = 0$$

Step 3: Plug in some values of x in $Q(x, y)$ and get the idea.

It isn't easy to plug in all values of x, but one can get the idea that it is not possible to make $Q(x, y)$ False.

In the above example, we can observe that $x.0$ is always 0 (zero) no matter what value of x could be plugged in.

d) $\exists x \exists y\, Q(x, y) \equiv \exists x \exists y\, (x.y = 0)$

"There exist some real numbers x and y such that x.y = 0"

Is it True? Yes.

Take $x = 1$, $y = 0$.

$x.y = 0$

$1.0 = 0$

Therefore, $\exists x \exists y\, Q(x, y)$ is true.

Nested Quantifiers (Example 3)

Example: Let $Q(x, y, z)$ be the statement "$x + y = z$." What are the truth values of the statements $\forall x \forall y \exists z\, Q(x, y, z)$ and $\exists z \forall x \forall y\, Q(x, y, z)$, where the domain of all variables consists of all real numbers?

Solution: (i) $\forall x \forall y \exists z\, Q(x, y, z) \equiv \forall x \forall y \exists z\, (x + y = z)$

"For all real numbers x and for all real numbers y, there is a real number z such that $x + y = z$" is true.

Let $x = 1$ and $y = -1$

$z = 0$

It is clear that for every combination of real numbers x and y, there is always a real number z such that $x + y = z$.

$\therefore \forall x \forall y \exists z\, (x + y = z)$ is True.

(ii) $\exists z \forall x \forall y\, Q(x, y, z) \equiv \exists z \forall x \forall y\, (x + y = z)$

"There exist some real number z such that for all real numbers x and y, $x + y = z$" is true.

Let $z = 2$

For $x = 1$ and $y = 1$, $x + y = z$ is true.

But, for $x = 2$ and $y = 3$, $x + y = z$ is not true.

It is clear that there is no real number z such that $x + y = z$ becomes true for all real numbers x and y.

$\therefore \exists z \forall x \forall y\, (x + y = z)$ is False.

Translating English Statements to Statement Involving Nested Quantifiers

It is not difficult to translate English statement to its equivalent logical expression if we follow the following steps:

Step 1: Read and understand the English statement given in the question.

Step 2: Find the domain of discourse according to the statement.

Step 3: Find the associated quantifiers.

Step 4: Rewrite the statement so that the domain of discourse and quantifiers are properly visible.

Step 5: Finally, introduce the variables in the statement.

Example 1: Translate the statement "The sum of two positive integers is always positive" into an equivalent logical expression.

Solution: English Statement

"The sum of two positive integers is always positive."

Step 1: Read and understand the statement.

Step 2: Find the domain of discourse.

Let us suppose that the domain consists of all integers.

Step 3: Find the associated quantifiers.

"The sum of two positive integers is always positive."

$\forall$ (for all) quantifier will be used.

Step 4: Rewrite the statement

Original statement -

"The sum of two positive integers is always positive."

Rewritten statement -

"For every two integers, if the two integers are positive then their sum is also positive."

Step 5: Introduce variables.

Two variables are required to represent two integers.
Let us suppose that integers are represented by the variables x and y.

"For every two integers x and y, if x and y are positive then x + y is also positive."

Equivalent logical expression: $\forall x \forall y\, ((x > 0) \land (y > 0) \rightarrow (x + y > 0))$

Translating English Statements to Statements involving Nested Quantifers

(Example 2)

Example 2: Translate the statement "A negative real number does not have a square root that is a real number" into a logical expression.

Solution: English statement: "A negative real number does not have a square root that is a real number."

Step 1: Read and understand the statement.

Step 2: Find the domain of discourse.

Let us suppose that domain consists of all real numbers.

Step 3: Find the associated quantifiers

→ All negative real numbers ($\forall$)

→ All negative real numbers have a square root which is not real ($\exists$)

Step 4: Rewrite the statement

Original statement - A negative real number does not have a square root that is a real number.

Rewritten statement - For all real numbers, if a real number is negative then its square root is not real.

Step 5: Introduce variables

For all real numbers x, if x is negative then it is not the case that there exist some real number y such that $x = y^2$.

Let us suppose y is a square root of x.

$$y = \sqrt{x} \Rightarrow y^2 = x$$

it is clear that if x is negative then there is no real number y such that $x = y^2$.

Equivalent logical expression: $\forall x\, ((x < 0) \rightarrow \neg\exists y\, (x = y^2))$

Negating Nested Quantifiers

Q1. What is the negation of the expression $\exists x\, P(x)$?

Ans. $\neg \exists x\, P(x) \equiv \forall x\, \neg P(x)$

Q2. What is the negation of $\forall x\, P(x)$?

Ans. $\neg \forall x\, P(x) \equiv \exists x\, \neg P(x)$

Negating nested quantifier is as simple as negating single quantifier.
We have to successively apply the rules of negating single quantifiers to nested quantifiers.

Example 1: Express the negation of the statement $\forall x \exists y\ (xy = 1)$ so that no negation precedes a quantifier.

Solution: Move the negation inside quantifiers by applying De Morgan's laws for negating single quantifiers.

According to De Morgan's law, $\forall$ will get converted to $\exists$ and $\exists$ will get converted to $\forall$.

$$\neg \forall x \exists y\ (xy = 1) \equiv \exists x \neg \exists y\ (xy = 1)$$

$$\exists x \neg \exists y\ (xy = 1) \equiv \exists x \forall y\ \neg(xy = 1)$$

$$\equiv \exists x \forall y\ (xy \neq 1)$$

Negating Nested Quantifiers

(Example 2)

Example 2: Express the statement "Someone has visited every country in the world except Libya" using quantifiers. Then form the negation of the statement so that no negation is to the left of a quantifier. Next, express the negation in simple English. (Do not simply use the words "It is not the case that.")

Solution: Let $V(x, y)$ denotes x has visited country y.
domain of x is "all people" and of y is "all countries."
"Someone has visited every country except Libya" is equivalent to

$$\exists x \forall y\, (V(x, y) \leftrightarrow y \neq \text{Libya})$$

Negation of the above statement:

$$\forall x \exists y\, (V(x, y) \leftrightarrow y = \text{Libya})$$

Remember:
$\neg(p \leftrightarrow q) \equiv (p \leftrightarrow \neg q)$

Every person has either visited Libya or has not visited any other country except Libya.

Negating Nested Quantifiers

(Example 3)

Example 3: Express the statement "No one has climbed every mountain in the Himalayas" using quantifiers. Then, form the negation of the statement so that no negation is to the left of a quantifier. Next, express the negation in simple English. (Do not use the words "It is not the case that.")

Solution: Let $C(x, y)$ denotes "x has climbed mountain y in the himalayas."

"No one has climbed every mountain in the Himalayas" can be translated to

$$\neg\exists x \forall y\, C(x, y)$$

Negation:

$$\neg(\neg\exists x \forall y\, C(x, y))$$

$$\equiv \exists x \forall y\, C(x, y)$$

There exist some person who has climbed every mountain in the Himalayas.

Exercise

1. Let $P(x)$ denote the statement "$x \leq 4$." What are these truth values?
   a) $P(0)$ b) $P(4)$ c) $P(6)$
2. Let $P(x)$ be the statement "the word x contains the letter a." What are these truth values?
   a) $P(\text{orange})$ b) $P(\text{lemon})$
   c) $P(\text{true})$ d) $P(\text{false})$
3. Let $Q(x, y)$ denote the statement "x is the capital of y." What are these truth values?
   a) $Q(\text{Denver, Colorado})$
   b) $Q(\text{Detroit, Michigan})$
   c) $Q(\text{Massachusetts, Boston})$
   d) $Q(\text{New York, New York})$
4. State the value of x after the statement **if** $P(x)$ **then** $x := 1$ is executed, where $P(x)$ is the statement "$x > 1$," if the value of x when this statement is reached is
   a) $x = 0$. b) $x = 1$.
   c) $x = 2$.
5. Let $P(x)$ be the statement "x spends more than five hours every weekday in class," where the domain for x consists of all students. Express each of these quantifications in English.
   a) $\exists x P(x)$ b) $\forall x P(x)$
   c) $\exists x \neg P(x)$ d) $\forall x \neg P(x)$
6. Let $N(x)$ be the statement "x has visited North Dakota," where the domain consists of the students in your school. Express each of these quantifications in English.
   a) $\exists x N(x)$ b) $\forall x N(x)$ c) $\neg\exists x N(x)$
   d) $\exists x \neg N(x)$ e) $\neg\forall x N(x)$ f) $\forall x \neg N(x)$
7. Translate these statements into English, where $C(x)$ is "x is a comedian" and $F(x)$ is "x is funny" and the domain consists of all people.
   a) $\forall x(C(x) \to F(x))$ b) $\forall x(C(x) \wedge F(x))$
   c) $\exists x(C(x) \to F(x))$ d) $\exists x(C(x) \wedge F(x))$
10. Let $C(x)$ be the statement "x has a cat," let $D(x)$ be the statement "x has a dog," and let $F(x)$ be the statement "x has a ferret." Express each of these statements in terms of $C(x)$, $D(x)$, $F(x)$, quantifiers, and logical connectives. Let the domain consist of all students in your class.
    a) A student in your class has a cat, a dog, and a ferret.
    b) All students in your class have a cat, a dog, or a ferret.
    c) Some student in your class has a cat and a ferret, but not a dog.
    d) No student in your class has a cat, a dog, and a ferret.
    e) For each of the three animals, cats, dogs, and ferrets, there is a student in your class who has this animal as a pet.
11. Let $P(x)$ be the statement "$x = x^2$." If the domain consists of the integers, what are these truth values?
    a) $P(0)$ b) $P(1)$ c) $P(2)$
    d) $P(-1)$ e) $\exists x P(x)$ f) $\forall x P(x)$
12. Let $Q(x)$ be the statement "$x + 1 > 2x$." If the domain consists of all integers, what are these truth values?
    a) $Q(0)$ b) $Q(-1)$ c) $Q(1)$
    d) $\exists x Q(x)$ e) $\forall x Q(x)$ f) $\exists x \neg Q(x)$
    g) $\forall x \neg Q(x)$
13. Determine the truth value of each of these statements if the domain consists of all integers.
    a) $\forall n(n + 1 > n)$ b) $\exists n(2n = 3n)$
    c) $\exists n(n = -n)$ d) $\forall n(3n \leq 4n)$
14. Determine the truth value of each of these statements if the domain consists of all real numbers.
    a) $\exists x(x^3 = -1)$ b) $\exists x(x^4 < x^2)$
    c) $\forall x((-x)^2 = x^2)$ d) $\forall x(2x > x)$
15. Determine the truth value of each of these statements if the domain for all variables consists of all integers.
    a) $\forall n(n^2 \geq 0)$ b) $\exists n(n^2 = 2)$
    c) $\forall n(n^2 \geq n)$ d) $\exists n(n^2 < 0)$

8. Translate these statements into English, where $R(x)$ is "x is a rabbit" and $H(x)$ is "x hops" and the domain consists of all animals.

a) $\forall x(R(x) \to H(x))$ b) $\forall x(R(x) \land H(x))$
c) $\exists x(R(x) \to H(x))$ d) $\exists x(R(x) \land H(x))$

9. Let $P(x)$ be the statement "x can speak Russian" and let $Q(x)$ be the statement "x knows the computer language C++." Express each of these sentences in terms of $P(x)$, $Q(x)$, quantifiers, and logical connectives. The domain for quantifiers consists of all students at your school.

a) There is a student at your school who can speak Russian and who knows C++.
b) There is a student at your school who can speak Russian but who doesn't know C++.
c) Every student at your school either can speak Russian or knows C++.
d) No student at your school can speak Russian or knows C++.

16. Determine the truth value of each of these statements if the domain of each variable consists of all real numbers.

a) $\exists x(x^2 = 2)$ b) $\exists x(x^2 = -1)$
c) $\forall x(x^2 + 2 \geq 1)$ d) $\forall x(x^2 \neq x)$

17. Suppose that the domain of the propositional function $P(x)$ consists of the integers 0, 1, 2, 3, and 4. Write out each of these propositions using disjunctions, conjunctions, and negations.

a) $\exists x P(x)$ b) $\forall x P(x)$ c) $\exists x \neg P(x)$
d) $\forall x \neg P(x)$ e) $\neg \exists x P(x)$ f) $\neg \forall x P(x)$

18. Suppose that the domain of the propositional function $P(x)$ consists of the integers $-2, -1, 0, 1$, and 2. Write out each of these propositions using disjunctions, conjunctions, and negations.

a) $\exists x P(x)$ b) $\forall x P(x)$ c) $\exists x \neg P(x)$
d) $\forall x \neg P(x)$ e) $\neg \exists x P(x)$ f) $\neg \forall x P(x)$

19. Suppose that the domain of the propositional function $P(x)$ consists of the integers 1, 2, 3, 4, and 5. Express these statements without using quantifiers, instead using only negations, disjunctions, and conjunctions.

a) $\exists x P(x)$ b) $\forall x P(x)$
c) $\neg \exists x P(x)$ d) $\neg \forall x P(x)$
e) $\forall x((x \neq 3) \to P(x)) \lor \exists x \neg P(x)$

20. Suppose that the domain of the propositional function $P(x)$ consists of $-5, -3, -1, 1, 3$, and 5. Express these statements without using quantifiers, instead using only negations, disjunctions, and conjunctions.

a) $\exists x P(x)$ b) $\forall x P(x)$
c) $\forall x((x \neq 1) \to P(x))$
d) $\exists x((x \geq 0) \land P(x))$
e) $\exists x(\neg P(x)) \land \forall x((x < 0) \to P(x))$

21. For each of these statements find a domain for which the statement is true and a domain for which the statement is false.

a) Everyone is studying discrete mathematics.
b) Everyone is older than 21 years.
c) Every two people have the same mother.
d) No two different people have the same grandmother.

22. For each of these statements find a domain for which the statement is true and a domain for which the statement is false.

a) Everyone speaks Hindi.
b) There is someone older than 21 years.
c) Every two people have the same first name.
d) Someone knows more than two other people.

23. Translate in two ways each of these statements into logical expressions using predicates, quantifiers, and logical connectives. First, let the domain consist of the students in your class and second, let it consist of all people.

a) Someone in your class can speak Hindi.
b) Everyone in your class is friendly.
c) There is a person in your class who was not born in California.

24. Translate in two ways each of these statements into logical expressions using predicates, quantifiers, and logical connectives. First, let the domain consist of the students in your class and second, let it consist of all people.

a) Everyone in your class has a cellular phone.
b) Somebody in your class has seen a foreign movie.
c) There is a person in your class who cannot swim.
d) All students in your class can solve quadratic equations.
e) Some student in your class does not want to be rich.

25. Translate each of these statements into logical expressions using predicates, quantifiers, and logical connectives.

a) No one is perfect.
b) Not everyone is perfect.
c) All your friends are perfect.
d) At least one of your friends is perfect.
e) Everyone is your friend and is perfect.
f) Not everybody is your friend or someone is not perfect.

26. Translate each of these statements into logical expressions in three different ways by varying the domain and by using predicates with one and with two variables.

a) Someone in your school has visited Uzbekistan.
b) Everyone in your class has studied calculus and C++.
c) No one in your school owns both a bicycle and a motorcycle.
d) There is a person in your school who is not happy.
e) Everyone in your school was born in the twentieth century.

27. Translate each of these statements into logical expressions in three different ways by varying the domain and by using predicates with one and with two variables.

a) A student in your school has lived in Vietnam.
b) There is a student in your school who cannot speak Hindi.
c) A student in your school knows Java, Prolog, and C++.
d) Everyone in your class enjoys Thai food.
e) Someone in your class does not play hockey.

28. Translate each of these statements into logical expressions using predicates, quantifiers, and logical connectives.

a) Something is not in the correct place.
b) All tools are in the correct place and are in excellent condition.
c) Everything is in the correct place and in excellent condition.
d) Nothing is in the correct place and is in excellent condition.
e) One of your tools is not in the correct place, but it is in excellent condition.

29. Express each of these statements using logical operators, predicates, and quantifiers.

a) Some propositions are tautologies.
b) The negation of a contradiction is a tautology.
c) The disjunction of two contingencies can be a tautology.
d) The conjunction of two tautologies is a tautology.

30. Suppose the domain of the propositional function $P(x, y)$ consists of pairs x and y, where x is 1, 2, or 3 and y is 1, 2, or 3. Write out these propositions using disjunctions and conjunctions.

a) $\exists x\, P(x, 3)$ **b)** $\forall y\, P(1, y)$

c) $\exists y \neg P(2, y)$ **d)** $\forall x\, \neg P(x, 2)$

31. Suppose that the domain of $Q(x, y, z)$ consists of triples x, y, z, where $x = 0, 1$, or 2, $y = 0$ or 1, and $z = 0$ or 1. Write out these propositions using disjunctions and conjunctions.

a) $\forall y\, Q(0, y, 0)$ **b)** $\exists x\, Q(x, 1, 1)$

c) $\exists z \neg Q(0, 0, z)$ **d)** $\exists x \neg Q(x, 0, 1)$

32. Express each of these statements using quantifiers. Then form the negation of the statement so that no negation is to the left of a quantifier. Next, express the negation in simple English. (Do not simply use the phrase "It is not the case that.")

a) All dogs have fleas.

b) There is a horse that can add.

c) Every koala can climb.

d) No monkey can speak French.

e) There exists a pig that can swim and catch fish.

33. Express each of these statements using quantifiers. Then form the negation of the statement, so that no negation is to the left of a quantifier. Next, express the negation in simple English. (Do not simply use the phrase "It is not the case that.")

a) Some old dogs can learn new tricks.

b) No rabbit knows calculus.

c) Every bird can fly.

d) There is no dog that can talk.

e) There is no one in this class who knows French and Russian.

34. Express the negation of these propositions using quantifiers, and then express the negation in English.

a) Some drivers do not obey the speed limit.

b) All Swedish movies are serious.

c) No one can keep a secret.

d) There is someone in this class who does not have a good attitude.

34. Express the negation of these propositions using quantifiers, and then express the negation in English.

a) Some drivers do not obey the speed limit.

b) All Swedish movies are serious.

c) No one can keep a secret.

d) There is someone in this class who does not have a good attitude.

35. Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all integers.

a) $\forall x(x^2 \geq x)$

b) $\forall x(x > 0 \vee x < 0)$

c) $\forall x(x = 1)$

36. Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all real numbers.

a) $\forall x(x^2 \neq x)$ **b)** $\forall x(x^2 \neq 2)$

c) $\forall x(|x| > 0)$

37. Express each of these statements using predicates and quantifiers.

a) A passenger on an airline qualifies as an elite flyer if the passenger flies more than 25,000 miles in a year or takes more than 25 flights during that year.

b) A man qualifies for the marathon if his best previous time is less than 3 hours and a woman qualifies for the marathon if her best previous time is less than 3.5 hours.

c) A student must take at least 60 course hours, or at least 45 course hours and write a master's thesis, and receive a grade no lower than a B in all required courses, to receive a master's degree.

d) There is a student who has taken more than 21 credit hours in a semester and received all A's.

Exercises 38–42 deal with the translation between system specification and logical expressions involving quantifiers.

38. Translate these system specifications into English where the predicate $S(x, y)$ is "x is in state y" and where the domain for x and y consists of all systems and all possible states, respectively.

a) $\exists x\, S(x, \text{open})$

b) $\forall x(S(x, \text{malfunctioning}) \vee S(x, \text{diagnostic}))$

c) $\exists x\, S(x, \text{open}) \vee \exists x\, S(x, \text{diagnostic})$

d) $\exists x \neg S(x, \text{available})$

e) $\forall x \neg S(x, \text{working})$

1. a) T, since $0 \leq 4$ **b)** T, since $4 \leq 4$ **c)** F, since $6 \not\leq 4$

3. a) This is true.

b) This is false, since Lansing, not Detroit, is the capital.

c) This is false (but $Q(\text{Boston}, \text{Massachusetts})$ is true).

d) This is false, since Albany, not New York, is the capital.

5. a) There is a student who spends more than five hours every weekday in class.

b) Every student spends more than five hours every weekday in class.

c) There is a student who does not spend more than five hours every weekday in class.

d) No student spends more than five hours every weekday in class. (Or, equivalently, every student spends less than or equal to five hours every weekday in class.)

7. a) This statement is that for every x, if x is a comedian, then x is funny. In English, this is most simply stated, "Every comedian is funny."

b) This statement is that for every x in the domain (universe of discourse), x is a comedian *and* x is funny. In English, this is most simply stated, "Every person is a funny comedian." Note that this is not the sort of thing one wants to say. It really makes no sense and doesn't say anything about the existence of boring comedians; it's surely false, because there exist lots of x for which $C(x)$ is false. This illustrates the fact that you rarely want to use conjunctions with universal quantifiers.

c) This statement is that there exists an x in the domain such that if x is a comedian then x is funny. In English, this might be rendered, "There exists a person such that if s/he is a comedian, then s/he is funny." Note that this is not the sort of thing one wants to say. It really makes no sense and doesn't say anything about the existence of funny comedians; it's surely true, because there exist lots of x for which $C(x)$ is false (recall the definition of the truth value of $p \to q$). This illustrates the fact that you rarely want to use conditional statements with existential quantifiers.

d) This statement is that there exists an x in the domain such that x is a comedian and x is funny. In English, this might be rendered, "There exists a funny comedian" or "Some comedians are funny" or "Some funny people are comedians."

9. **a)** We assume that this sentence is asserting that the same person has both talents. Therefore we can write $\exists x(P(x) \wedge Q(x))$.

b) Since "but" really means the same thing as "and" logically, this is $\exists x(P(x) \wedge \neg Q(x))$

c) This time we are making a universal statement: $\forall x(P(x) \vee Q(x))$

d) This sentence is asserting the nonexistence of anyone with either talent, so we could write it as $\neg\exists x(P(x) \vee Q(x))$. Alternatively, we can think of this as asserting that everyone fails to have either of these talents, and we obtain the logically equivalent answer $\forall x\neg(P(x) \vee Q(x))$. Failing to have either talent is equivalent to having neither talent (by De Morgan's law), so we can also write this as $\forall x((\neg P(x)) \wedge (\neg Q(x))$. Note that it would *not* be correct to write $\forall x((\neg P(x)) \vee (\neg Q(x))$ nor to write $\forall x\neg(P(x) \wedge Q(x))$.

11. **a)** T, since $0 = 0^2$ **b)** T, since $1 = 1^2$ **c)** F, since $2 \neq 2^2$
d) F, since $-1 \neq (-1)^2$ **e)** T (let $x = 1$) **f)** F (let $x = 2$)

13. **a)** Since adding 1 to a number makes it larger, this is true.

b) Since $2 \cdot 0 = 3 \cdot 0$, this is true.

c) This statement is true, since $0 = -0$.

d) This is true for the nonnegative integers but not for the negative integers. For example, $3(-2) \not\leq 4(-2)$. Therefore the universally quantified statement is false.

15. Recall that the integers include the positive and negative integers and 0.

a) This is the well-known true fact that the square of a real number cannot be negative.

b) There are two *real* numbers that satisfy $n^2 = 2$, namely $\pm\sqrt{2}$, but there do not exist any *integers* with this property, so the statement is false.

c) If n is a positive integer, then $n^2 \geq n$ is certainly true; it's also true for $n = 0$; and it's trivially true if n is negative. Therefore the universally quantified statement is true.

d) Squares can never be negative; therefore this statement is false.

17. Existential quantifiers are like disjunctions, and universal quantifiers are like conjunctions. See Examples 11 and 16.

a) We want to assert that $P(x)$ is true for some x in the universe, so either $P(0)$ is true or $P(1)$ is true or $P(2)$ is true or $P(3)$ is true or $P(4)$ is true. Thus the answer is $P(0) \vee P(1) \vee P(2) \vee P(3) \vee P(4)$. The other parts of this exercise are similar. Note that by De Morgan's laws, the expression in part **(c)** is logically equivalent to the expression in part **(f)**, and the expression in part **(d)** is logically equivalent to the expression in part **(e)**.

b) $P(0) \wedge P(1) \wedge P(2) \wedge P(3) \wedge P(4)$

c) $\neg P(0) \vee \neg P(1) \vee \neg P(2) \vee \neg P(3) \vee \neg P(4)$

d) $\neg P(0) \wedge \neg P(1) \wedge \neg P(2) \wedge \neg P(3) \wedge \neg P(4)$

e) This is just the negation of part **(a)**: $\neg(P(0) \vee P(1) \vee P(2) \vee P(3) \vee P(4))$

f) This is just the negation of part **(b)**: $\neg(P(0) \wedge P(1) \wedge P(2) \wedge P(3) \wedge P(4))$

19. Existential quantifiers are like disjunctions, and universal quantifiers are like conjunctions. See Examples 11 and 16.

a) We want to assert that $P(x)$ is true for some x in the universe, so either $P(1)$ is true or $P(2)$ is true or $P(3)$ is true or $P(4)$ is true or $P(5)$ is true. Thus the answer is $P(1) \vee P(2) \vee P(3) \vee P(4) \vee P(5)$.

b) $P(1) \wedge P(2) \wedge P(3) \wedge P(4) \wedge P(5)$

c) This is just the negation of part **(a)**: $\neg(P(1) \vee P(2) \vee P(3) \vee P(4) \vee P(5))$

d) This is just the negation of part **(b)**: $\neg(P(1) \wedge P(2) \wedge P(3) \wedge P(4) \wedge P(5))$

e) The formal translation is as follows: $(((1 \neq 3) \rightarrow P(1)) \wedge ((2 \neq 3) \rightarrow P(2)) \wedge ((3 \neq 3) \rightarrow P(3)) \wedge ((4 \neq 3) \rightarrow P(4)) \wedge ((5 \neq 3) \rightarrow P(5))) \vee (\neg P(1) \vee \neg P(2) \vee \neg P(3) \vee \neg P(4) \vee \neg P(5))$. However, since the hypothesis $x \neq 3$ is false when x is 3 and true when x is anything other than 3, we have more simply $(P(1) \wedge P(2) \wedge P(4) \wedge P(5)) \vee (\neg P(1) \vee \neg P(2) \vee \neg P(3) \vee \neg P(4) \vee \neg P(5))$. Thinking about it a little more, we note that this statement is always true, since if the first part is not true, then the second part must be true.

21. **a)** One would hope that if we take the domain to be the students in your class, then the statement is true. If we take the domain to be all students in the world, then the statement is clearly false, because some of them are studying only other subjects.

b) If we take the domain to be United States Senators, then the statement is true. If we take the domain to be college football players, then the statement is false, because some of them are younger than 21.

c) If the domain consists of just Princes William and Harry of Great Britain (sons of the late Princess Diana), then the statement is true. It is also true if the domain consists of just one person (everyone has the same mother as him- or herself). If the domain consists of all the grandchildren of Queen Elizabeth II of Great Britain (of whom William and Harry are just two), then the statement is false.

d) If the domain consists of Bill Clinton and George W. Bush, then this statement is true because they do not have the same grandmother. If the domain consists of all residents of the United States, then the statement is false, because there are many instances of siblings and first cousins, who have at least one grandmother in common.

23. In order to do the translation the second way, we let $C(x)$ be the propositional function "x is in your class." Note that for the second way, we always want to use conditional statements with universal quantifiers and conjunctions with existential quantifiers.

a) Let $H(x)$ be "x can speak Hindi." Then we have $\exists x\, H(x)$ the first way, or $\exists x(C(x) \land H(x))$ the second way.

b) Let $F(x)$ be "x is friendly." Then we have $\forall x\, F(x)$ the first way, or $\forall x(C(x) \to F(x))$ the second way.

c) Let $B(x)$ be "x was born in California." Then we have $\exists x\, \neg B(x)$ the first way, or $\exists x(C(x) \land \neg B(x))$ the second way.

d) Let $M(x)$ be "x has been in a movie." Then we have $\exists x\, M(x)$ the first way, or $\exists x(C(x) \land M(x))$ the second way.

e) This is saying that everyone has failed to take the course. So the answer here is $\forall x\, \neg L(x)$ the first way, or $\forall x(C(x) \to \neg L(x))$ the second way, where $L(x)$ is "x has taken a course in logic programming."

25. Let $P(x)$ be "x is perfect"; let $F(x)$ be "x is your friend"; and let the domain (universe of discourse) be all people.

a) This means that everyone has the property of being not perfect: $\forall x \neg P(x)$. Alternatively, we can write this as $\neg\exists x\, P(x)$, which says that there does not exist a perfect person.

b) This is just the negation of "Everyone is perfect": $\neg\forall x\, P(x)$.

c) If someone is your friend, then that person is perfect: $\forall x(F(x) \rightarrow P(x))$. Note the use of conditional statement with universal quantifiers.

d) We do not have to rule out your having more than one perfect friend. Thus we have simply $\exists x(F(x) \wedge P(x))$. Note the use of conjunction with existential quantifiers.

e) The expression is $\forall x(F(x) \wedge P(x))$. Note that here we did use a conjunction with the universal quantifier, but the sentence is not natural (who could claim this?). We could also have split this up into two quantified statements and written $(\forall x\, F(x)) \wedge (\forall x\, P(x))$.

f) This is a disjunction. The expression is $(\neg\forall x\, F(x)) \vee (\exists x \neg P(x))$.

27. In all of these, we will let $Y(x)$ be the propositional function that x is in your school or class, as appropriate.

a) If we let $V(x)$ be "x has lived in Vietnam," then we have $\exists x\, V(x)$ if the universe is just your schoolmates, or $\exists x(Y(x) \wedge V(x))$ if the universe is all people. If we let $D(x, y)$ mean that person x has lived in country y, then we can rewrite this last one as $\exists x(Y(x) \wedge D(x, \text{Vietnam}))$.

b) If we let $H(x)$ be "x can speak Hindi," then we have $\exists x \neg H(x)$ if the universe is just your schoolmates, or $\exists x(Y(x) \wedge \neg H(x))$ if the universe is all people. If we let $S(x, y)$ mean that person x can speak language y, then we can rewrite this last one as $\exists x(Y(x) \wedge \neg S(x, \text{Hindi}))$.

c) If we let $J(x)$, $P(x)$, and $C(x)$ be the propositional functions asserting x's knowledge of Java, Prolog, and C++, respectively, then we have $\exists x(J(x) \wedge P(x) \wedge C(x))$ if the universe is just your schoolmates, or $\exists x(Y(x) \wedge J(x) \wedge P(x) \wedge C(x))$ if the universe is all people. If we let $K(x, y)$ mean that person x knows programming language y, then we can rewrite this last one as $\exists x(Y(x) \wedge K(x, \text{Java}) \wedge K(x, \text{Prolog}) \wedge K(x, \text{C++}))$.

d) If we let $T(x)$ be "x enjoys Thai food," then we have $\forall x\, T(x)$ if the universe is just your classmates, or $\forall x(Y(x) \rightarrow T(x))$ if the universe is all people. If we let $E(x, y)$ mean that person x enjoys food of type y, then we can rewrite this last one as $\forall x(Y(x) \rightarrow E(x, \text{Thai}))$.

e) If we let $H(x)$ be "x plays hockey," then we have $\exists x \neg H(x)$ if the universe is just your classmates, or $\exists x(Y(x) \wedge \neg H(x))$ if the universe is all people. If we let $P(x, y)$ mean that person x plays game y, then we can rewrite this last one as $\exists x(Y(x) \wedge \neg P(x, \text{hockey}))$.

29. Our domain (universe of discourse) here is all propositions. Let $T(x)$ mean that x is a tautology and $C(x)$ mean that x is a contradiction. Since a contingency is just a proposition that is neither a tautology nor a contradiction, we do not need a separate predicate for being a contingency.

a) This one is just the assertion that tautologies exist: $\exists x\, T(x)$.

b) Although the word "all" or "every" does not appear here, this sentence is really expressing a universal meaning, that the negation of a contradiction is always a tautology. So we want to say that if x is a contradiction, then $\neg x$ is a tautology. Thus we have $\forall x(C(x) \rightarrow T(\neg x))$. Note the rare use of a logical symbol (negation) applied to a variable (x); this is purely a coincidence in this exercise because the universe happens itself to be propositions.

c) The words "can be" are expressing an existential idea—that there exist two contingencies whose disjunction is a tautology. Thus we have $\exists x \exists y(\neg T(x) \wedge \neg C(x) \wedge \neg T(y) \wedge \neg C(y) \wedge T(x \vee y))$. The same final comment as in part **(b)** applies here. Also note the explanation about contingencies in the preamble.

d) As in part **(b)**, this is the universal assertion that whenever x and y are tautologies, then so is $x \wedge y$; thus we have $\forall x \forall y((T(x) \wedge T(y)) \rightarrow T(x \wedge y))$.

31. In each case we just have to list all the possibilities, joining them with $\vee$ if the quantifier is $\exists$, and joining them with $\wedge$ if the quantifier is $\forall$.

a) $Q(0,0,0) \wedge Q(0,1,0)$ **b)** $Q(0,1,1) \vee Q(1,1,1) \vee Q(2,1,1)$

c) $\neg Q(0,0,0) \vee \neg Q(0,0,1)$ **d)** $\neg Q(0,0,1) \vee \neg Q(1,0,1) \vee \neg Q(2,0,1)$

33. In each case we need to specify some predicates and identify the domain (universe of discourse).

a) Let $T(x)$ be the predicate that x can learn new tricks, and let the domain be old dogs. Our original statement is $\exists x\,T(x)$. Its negation is $\neg\exists x\,T(x)$, which we must to rewrite in the required manner as $\forall x\,\neg T(x)$.

In English this reads "Every old dog is unable to learn new tricks" or "All old dogs can't learn new tricks." (Note that this does *not* say that not all old dogs can learn new tricks—it is saying something stronger than that.) More colloquially, we can say "No old dogs can learn new tricks."

b) Let $C(x)$ be the predicate that x knows calculus, and let the domain be rabbits. Our original statement is $\neg\exists x\,C(x)$. Its negation is, of course, simply $\exists x\,C(x)$. In English this reads "There is a rabbit that knows calculus."

c) Let $F(x)$ be the predicate that x can fly, and let the domain be birds. Our original statement is $\forall x\,F(x)$. Its negation is $\neg\forall x\,F(x)$ (i.e., not all birds can fly), which we must to rewrite in the required manner as $\exists x\,\neg F(x)$. In English this reads "There is a bird who cannot fly."

d) Let $T(x)$ be the predicate that x can talk, and let the domain be dogs. Our original statement is $\neg\exists x\,T(x)$. Its negation is, of course, simply $\exists x\,T(x)$. In English this reads "There is a dog that talks."

e) Let $F(x)$ and $R(x)$ be the predicates that x knows French and knows Russian, respectively, and let the domain be people in this class. Our original statement is $\neg\exists x(F(x)\wedge R(x))$. Its negation is, of course, simply $\exists x(F(x)\wedge R(x))$. In English this reads "There is someone in this class who knows French and Russian."

35. **a)** As we saw in Example 13, this is true, so there is no counterexample.

b) Since 0 is neither greater than nor less than 0, this is a counterexample.

c) This proposition says that 1 is the only integer—that every integer equals 1. If is obviously false, and any other integer, such as -111749, provides a counterexample.

37. In each case we need to make up predicates. The answers are certainly not unique and depend on the choice of predicate, among other things.

a) $\forall x((F(x,25000)\vee S(x,25))\rightarrow E(x))$, where $E(x)$ is "Person x qualifies as an elite flyer in a given year," $F(x,y)$ is "Person x flies more than y miles in a given year," and $S(x,y)$ is "Person x takes more than y flights in a given year"

b) $\forall x(((M(x)\wedge T(x,3))\vee(\neg M(x)\wedge T(x,3.5)))\rightarrow Q(x))$, where $Q(x)$ is "Person x qualifies for the marathon," $M(x)$ is "Person x is a man," and $T(x,y)$ is "Person x has run the marathon in less than y hours"

c) $M\rightarrow((H(60)\vee(H(45)\wedge T))\wedge\forall y\,G(\mathrm{B},y))$, where M is the proposition "The student received a masters degree," $H(x)$ is "The student took at least x course hours," T is the proposition "The student wrote a thesis," and $G(x,y)$ is "The person got grade x or higher in his course y"

d) $\exists x\,((T(x,21)\wedge G(x,4.0))$, where $T(x,y)$ is "Person x took more than y credit hours" and $G(x,p)$ is "Person x earned grade point average p" (we assume that we are talking about one given semester)

39. In each case we pretty much just write what we see.

a) If there is a printer that is both out of service and busy, then some job has been lost.

b) If every printer is busy, then there is a job in the queue.

c) If there is a job that is both queued and lost, then some printer is out of service.

d) If every printer is busy and every job is queued, then some job is lost.

Exercises

1. Translate these statements into English, where the domain for each variable consists of all real numbers.
   a) $\forall x \exists y(x < y)$
   b) $\forall x \forall y(((x \geq 0) \land (y \geq 0)) \to (xy \geq 0))$
   c) $\forall x \forall y \exists z(xy = z)$
2. Translate these statements into English, where the domain for each variable consists of all real numbers.
   a) $\exists x \forall y(xy = y)$
   b) $\forall x \forall y(((x \geq 0) \land (y < 0)) \to (x - y > 0))$
   c) $\forall x \forall y \exists z(x = y + z)$
3. Let $Q(x, y)$ be the statement "x has sent an e-mail message to y," where the domain for both x and y consists of all students in your class. Express each of these quantifications in English.
   a) $\exists x \exists y Q(x, y)$
   b) $\exists x \forall y Q(x, y)$
   c) $\forall x \exists y Q(x, y)$
   d) $\exists y \forall x Q(x, y)$
   e) $\forall y \exists x Q(x, y)$
   f) $\forall x \forall y Q(x, y)$
4. Let $P(x, y)$ be the statement "Student x has taken class y," where the domain for x consists of all students in your class and for y consists of all computer science courses at your school. Express each of these quantifications in English.
   a) $\exists x \exists y P(x, y)$
   b) $\exists x \forall y P(x, y)$
   c) $\forall x \exists y P(x, y)$
   d) $\exists y \forall x P(x, y)$
   e) $\forall y \exists x P(x, y)$
   f) $\forall x \forall y P(x, y)$
5. Let $W(x, y)$ mean that student x has visited website y, where the domain for x consists of all students in your school and the domain for y consists of all websites. Express each of these statements by a simple English sentence.
   a) W(Sarah Smith, www.att.com)
   b) $\exists x W(x,$ www.imdb.org)
   c) $\exists y W$(José Orez, y)
   d) $\exists y(W$(Ashok Puri, $y) \land W$(Cindy Yoon, y))
   e) $\exists y \forall z(y \neq$ (David Belcher) $\land$ (W(David Belcher, z) $\to W(y,z)$))
   f) $\exists x \exists y \forall z((x \neq y) \land (W(x, z) \leftrightarrow W(y, z)))$

7. Let $T(x, y)$ mean that student x likes cuisine y, where the domain for x consists of all students at your school and the domain for y consists of all cuisines. Express each of these statements by a simple English sentence.
   a) $\neg T$(Abdallah Hussein, Japanese)
   b) $\exists x T(x,$ Korean) $\land \forall x T(x,$ Mexican)
   c) $\exists y(T$(Monique Arsenault, y) $\lor$ T(Jay Johnson, y))
   d) $\forall x \forall z \exists y((x \neq z) \to \neg(T(x, y) \land T(z, y)))$
   e) $\exists x \exists z \forall y(T(x, y) \leftrightarrow T(z, y))$
   f) $\forall x \forall z \exists y(T(x, y) \leftrightarrow T(z, y))$

1. a) For every real number x there exists a real number y such that x is less than y. Basically, this is asserting that there is no largest real number—for any real number you care to name, there is a larger one.

b) For every real number x and real number y, if x and y are both nonnegative, then their product is nonnegative. Or, more simply, the product of nonnegative real numbers is nonnegative.

c) For every real number x and real number y, there exists a real number z such that $xy = z$. Or, more simply, the real numbers are closed under multiplication. (Some authors would include the uniqueness of z as part of the meaning of the word *closed*.)

3. It is useful to keep in mind that x and y can be the same person, so sending messages to oneself counts in this problem.

a) Formally, this says that there exist students x and y such that x has sent a message to y. In other words, there is some student in your class who has sent a message to some student in your class.

b) This is similar to part **(a)** except that x has sent a message to everyone, not just to at least one person. So this says there is some student in your class who has sent a message to every student in your class.

c) Note that this is not the same as part **(b)**. Here we have that for every x there exists a y such that x has sent a message to y. In other words, every student in your class has sent a message to at least one student in your class.

d) Note that this is not the same as part **(c)**, since the order of quantifiers has changed. In part **(c)**, y could depend on x; in other words, the recipient of the messages could vary from sender to sender. Here the existential quantification on y comes first, so it's the same recipient for all the messages. The meaning is that there is a student in your class who has been sent a message by every student in your class.

e) This is similar to part **(c)**, with the role of sender and recipient reversed: every student in your class has been sent a message from at least one student in your class. Again, note that the sender can depend on the recipient.

f) Every student in the class has sent a message to every student in the class.

7. **a)** Abdallah Hussein does not like Japanese cuisine.

b) Note that this is the conjunction of two separate quantified statements. Some student at your school likes Korean cuisine, and everyone at your school likes Mexican cuisine.

c) There is some cuisine that either Monique Arsenault or Jay Johnson likes.

d) Formally this says that for every x and z, there exists a y such that if x and z are not equal, then it is not the case that both x and z like y. In simple English, this says that for every pair of distinct students at

your school, there is some cuisine that at least one them does not like.

e) There are two students at your school who have exactly the same tastes (i.e., they like exactly the same cuisines).

f) For every pair of students at your school, there is some cuisine about which they have the same opinion (either they both like it or they both do not like it).

9. Let $L(x, y)$ be the statement "x loves y," where the domain for both x and y consists of all people in the world. Use quantifiers to express each of these statements.

a) Everybody loves Jerry.
b) Everybody loves somebody.
c) There is somebody whom everybody loves.
d) Nobody loves everybody.
e) There is somebody whom Lydia does not love.
f) There is somebody whom no one loves.
g) There is exactly one person whom everybody loves.
h) There are exactly two people whom Lynn loves.
i) Everyone loves himself or herself.
j) There is someone who loves no one besides himself or herself.

9. We need to be careful to put the lover first and the lovee second as arguments in the propositional function L.

a) $\forall x L(x, \text{Jerry})$

b) Note that the "somebody" being loved depends on the person doing the loving, so we have to put the universal quantifier first: $\forall x \exists y L(x, y)$.

c) In this case, one lovee works for all lovers, so we have to put the existential quantifier first: $\exists y \forall x L(x, y)$.

d) We could think of this as saying that there does not exist anyone who loves everybody ($\neg \exists x \forall y L(x, y)$), or we could think of it as saying that for each person, we can find a person whom he or she does not love ($\forall x \exists y \neg L(x, y)$. These two expressions are logically equivalent.

e) $\exists x \neg L(\text{Lydia}, x)$

f) We are asserting the existence of an individual such that everybody fails to love that person: $\exists x \forall y \neg L(y, x)$.

g) In Exercises 52–54 of Section 1.4, we worked with a notation for the existence of a unique object satisfying a certain condition. Employing that device, we could write this as $\exists! x \forall y L(y, x)$. In Exercise 52 of the present section we will discover a way to avoid this notation in general. What we have to say is that the x asserted here exists, and that every z satisfying this condition (of being loved by everybody) must equal x. Thus we obtain $\exists x \big(\forall y L(y, x) \land \forall z ((\forall w L(w, z)) \to z = x)\big)$. Note that we could have used y as the bound variable where we used w; since the scope of the first use of y had ended before we came to this point in the formula, reusing y as the bound variable would cause no ambiguity.

h) We want to assert the existence of two distinct people, whom we will call x and y, whom Lynn loves, as well as make the statement that everyone whom Lynn loves must be either x or y: $\exists x \exists y \big(x \neq y \land L(\text{Lynn}, x) \land L(\text{Lynn}, y) \land \forall z (L(\text{Lynn}, z) \to (z = x \lor z = y))\big)$.

i) $\forall x L(x, x)$ (Note that nothing in our earlier answers ruled out the possibility that variables or constants with different names might be equal to each other. For example, in part (**a**), x could equal Jerry, so that statement includes as a special case the assertion that Jerry loves himself. Similarly, in part (**h**), the two people whom Lynn loves either could be two people other than Lynn (in which case we know that Lynn does not love herself), or could be Lynn herself and one other person.)

j) This is asserting that the one and only one person who is loved by the person being discussed is in fact that person: $\exists x \forall y (L(x, y) \leftrightarrow x = y)$.

11. Let $S(x)$ be the predicate "x is a student," $F(x)$ the predicate "x is a faculty member," and $A(x, y)$ the predicate "x has asked y a question," where the domain consists of all people associated with your school. Use quantifiers to express each of these statements.

a) Lois has asked Professor Michaels a question.

b) Every student has asked Professor Gross a question.

c) Every faculty member has either asked Professor Miller a question or been asked a question by Professor Miller.

d) Some student has not asked any faculty member a question.

e) There is a faculty member who has never been asked a question by a student.

f) Some student has asked every faculty member a question.

g) There is a faculty member who has asked every other faculty member a question.

h) Some student has never been asked a question by a faculty member.

11. **a)** We might want to assert that Lois is a student and Michaels is a faculty member, but the sentence doesn't really say that, so the simple answer is just $A(\text{Lois}, \text{Professor Michaels})$.

b) To say that every student (as opposed to every person) has done this, we need to restrict our universally quantified variable to being a student. The easiest way to do this is to make the assertion being quantified a conditional statement. *As a general rule of thumb, use conditional statements with universal quantifiers and conjunctions with existential quantifiers (see part* **(d)***, for example).* Thus our answer is $\forall x(S(x) \to A(x, \text{Professor Gross}))$.

c) This is similar to part **(b)**: $\forall x(F(x) \to (A(x, \text{Professor Miller}) \lor A(\text{Professor Miller}, x)))$. Note the need for parentheses in these answers.

d) There is a student such that for every faculty member, that student has not asked that faculty member a question. Note how we need to include the S and F predicates: $\exists x(S(x) \land \forall y(F(y) \to \neg A(x, y)))$. We could also write this as $\exists x(S(x) \land \neg \exists y(F(y) \land A(x, y)))$.

e) This is very similar to part **(d)**, with the role of the players reversed: $\exists x(F(x) \land \forall y(S(y) \to \neg A(y, x)))$.

f) This is a little ambiguous in English. If the statement is that there is a very inquisitive student, one who has gone around and asked a question of every professor, then this is similar to part **(d)**, without the negation: $\exists x(S(x) \land \forall y(F(y) \to A(x, y)))$. On the other hand, the statement might be intended as asserting simply that for every professor, there exists some student who has asked that professor a question. In other words, the questioner might depend on the questionee. Note how the meaning changes with the change in order of quantifiers. Under the second interpretation the answer is $\forall y(F(y) \to \exists x(S(x) \land A(x, y)))$. The first interpretation is probably the intended one.

g) This is pretty straightforward, except that we have to rule out the possibility that the askee is the same as the asker. Our sentence needs to say that there exists a faculty member such that for every other faculty member, the first has asked the second a question: $\exists x(F(x) \land \forall y((F(y) \land y \neq x) \to A(x, y)))$.

h) There is a student such that every faculty member has failed to ask him a question: $\exists x(S(x) \land \forall y(F(y) \to \neg A(y, x)))$.

13. Let $M(x, y)$ be "x has sent y an e-mail message" and $T(x, y)$ be "x has telephoned y," where the domain consists of all students in your class. Use quantifiers to express each of these statements. (Assume that all e-mail messages that were sent are received, which is not the way things often work.)

a) Chou has never sent an e-mail message to Koko.

b) Arlene has never sent an e-mail message to or telephoned Sarah.

c) José has never received an e-mail message from Deborah.

d) Every student in your class has sent an e-mail message to Ken.

e) No one in your class has telephoned Nina.

f) Everyone in your class has either telephoned Avi or sent him an e-mail message.

g) There is a student in your class who has sent everyone else in your class an e-mail message.

h) There is someone in your class who has either sent an e-mail message or telephoned everyone else in your class.

i) There are two different students in your class who have sent each other e-mail messages.

j) There is a student who has sent himself or herself an e-mail message.

k) There is a student in your class who has not received an e-mail message from anyone else in the class and who has not been called by any other student in the class.

l) Every student in the class has either received an e-mail message or received a telephone call from another student in the class.

m) There are at least two students in your class such that one student has sent the other e-mail and the second student has telephoned the first student.

n) There are two different students in your class who between them have sent an e-mail message to or telephoned everyone else in the class.

13. Be careful to put in parentheses where needed; otherwise your answer can be either ambiguous or wrong.

a) Clearly this is simply $\neg M(\text{Chou}, \text{Koko})$.

b) We can give two answers, which are equivalent by De Morgan's law: $\neg(M(\text{Arlene}, \text{Sarah}) \vee T(\text{Arlene}, \text{Sarah}))$ or $\neg M(\text{Arlene}, \text{Sarah}) \wedge \neg T(\text{Arlene}, \text{Sarah})$.

c) Clearly this is simply $\neg M(\text{Deborah}, \text{Jose})$.

d) Note that this statement includes the assertion that Ken has sent himself a message: $\forall x\, M(x, \text{Ken})$.

e) We can write this in two equivalent ways, depending on whether we want to say that everyone has failed to phone Nina or to say that there does not exist someone who has phoned her: $\forall x\, \neg T(x, \text{Nina})$ or $\neg\exists x\, T(x, \text{Nina})$.

f) This is almost identical to part **(d)**: $\forall x(T(x, \text{Avi}) \vee M(x, \text{Avi}))$.

g) To get the "else" in there, we have to make sure that y is different from x in our answer: $\exists x \forall y(y \neq x \rightarrow M(x, y))$.

h) This is almost identical to part **(g)**: $\exists x \forall y(y \neq x \rightarrow (M(x, y) \vee T(x, y)))$.

i) We need to assert the existence of two distinct people who have sent e-mail both ways: $\exists x \exists y(x \neq y \wedge M(x, y) \wedge M(y, x))$.

j) Only one variable is needed: $\exists x\, M(x, x)$.

k) This poor soul (x in our expression) did not receive a message or a phone call (i.e., did not receive a message and did not receive a phone call) from any person y other than possibly himself: $\exists x \forall y(x \neq y \rightarrow (\neg M(y, x) \wedge \neg T(y, x)))$.

l) Here y is "another student": $\forall x \exists y(x \neq y \wedge (M(y, x) \vee T(y, x)))$.

m) This is almost identical to part **(i)**: $\exists x \exists y(x \neq y \wedge M(x, y) \wedge T(y, x))$.

n) Note how the "everyone else" means someone different from both x and y in our expression (and note that there are four possibilities for how each such person z might be contacted): $\exists x \exists y(x \neq y \wedge \forall z((z \neq x \wedge z \neq y) \rightarrow (M(x, z) \vee M(y, z) \vee T(x, z) \vee T(y, z))))$.

25. Translate each of these nested quantifications into an English statement that expresses a mathematical fact. The domain in each case consists of all real numbers.

a) $\exists x \forall y(xy = y)$
b) $\forall x \forall y(((x < 0) \land (y < 0)) \rightarrow (xy > 0))$
c) $\exists x \exists y((x^2 > y) \land (x < y))$
d) $\forall x \forall y \exists z(x + y = z)$

25. a) This says that there exists a real number x such that for every real number y, the product xy equals y. That is, there is a multiplicative identity for the real numbers. This is a true statement, since $x = 1$ is the identity.

b) The product of two negative real numbers is always a positive real number.

c) There exist real numbers x and y such that x^2 exceeds y but x is less than y. This is true, since we can take $x = 2$ and $y = 3$, for instance.

d) This says that for every pair of real numbers x and y, there exists a real number z that is their sum. In other words, the real numbers are closed under the operation of addition, another true fact. (Some authors would include the uniqueness of z as part of the meaning of the word *closed*.)

27. Determine the truth value of each of these statements if the domain for all variables consists of all integers.

a) $\forall n \exists m (n^2 < m)$ **b)** $\exists n \forall m (n < m^2)$
c) $\forall n \exists m (n + m = 0)$ **d)** $\exists n \forall m (nm = m)$

27. Recall that the integers include the positive and negative integers and 0.

a) The import of this statement is that no matter how large n might be, we can always find an integer m bigger than n^2. This is certainly true; for example, we could always take $m = n^2 + 1$.

b) This statement is asserting that there is an n that is smaller than the square of *every* integer; note that n is not allowed to depend on m, since the existential quantifier comes first. This statement is true, since we could take, for instance, $n = -3$, and then n would be less than every square, since squares are always greater than or equal to 0.

c) Note the order of quantifiers: m here is allowed to depend on n. Since we can take $m = -n$, this statement is true (additive inverses exist for the integers).

d) Here one n must work for all m. Clearly $n = 1$ does the trick, so the statement is true.

e) The statement is that the equation $n^2 + m^2 = 5$ has a solution over the integers. This is true; in fact there are eight solutions, namely $n = \pm 1$, $m = \pm 2$, and vice versa.

f) The statement is that the equation $n^2 + m^2 = 6$ has a solution over the integers. There are only a small finite number of cases to try, since if $|m|$ or $|n|$ were bigger than 2 then the left-hand side would be bigger than 6. A few minutes reflection shows that in fact there is no solution, so the existential statement is false.

g) The statement is that the system of equations $\{n + m = 4, n - m = 1\}$ has a solution over the integers. By algebra we see that there is a unique solution to this system, namely $n = 2\frac{1}{2}$, $m = 1\frac{1}{2}$. Since there do not exist *integers* that make the equations true, the statement is false.

h) The statement is that the system of equations $\{n + m = 4, n - m = 2\}$ has a solution over the integers. By algebra we see that there is indeed an integral solution to this system, namely $n = 3$, $m = 1$. Therefore the statement is true.

i) This statement says that the average of two integers is always an integer. If we take $m = 1$ and $n = 2$, for example, then the only p for which $p = (m + n)/2$ is $p = 1\frac{1}{2}$, which is not an integer. Therefore the statement is false.

29. Suppose the domain of the propositional function $P(x, y)$ consists of pairs x and y, where x is 1, 2, or 3 and y is 1, 2, or 3. Write out these propositions using disjunctions and conjunctions.

a) $\forall x \forall y P(x, y)$

b) $\exists x \exists y P(x, y)$

c) $\exists x \forall y P(x, y)$

d) $\forall y \exists x P(x, y)$

29. **a)** $P(1,1) \wedge P(1,2) \wedge P(1,3) \wedge P(2,1) \wedge P(2,2) \wedge P(2,3) \wedge P(3,1) \wedge P(3,2) \wedge P(3,3)$

b) $P(1,1) \vee P(1,2) \vee P(1,3) \vee P(2,1) \vee P(2,2) \vee P(2,3) \vee P(3,1) \vee P(3,2) \vee P(3,3)$

c) $(P(1,1) \wedge P(1,2) \wedge P(1,3)) \vee (P(2,1) \wedge P(2,2) \wedge P(2,3)) \vee (P(3,1) \wedge P(3,2) \wedge P(3,3))$

d) $(P(1,1) \vee P(2,1) \vee P(3,1)) \wedge (P(1,2) \vee P(2,2) \vee P(3,2)) \wedge (P(1,3) \vee P(2,3) \vee P(3,3))$

Note the crucial difference between parts **(c)** and **(d)**.

31. Express the negations of each of these statements so that all negation symbols immediately precede predicates.

a) $\forall x \exists y \forall z T(x, y, z)$

b) $\forall x \exists y P(x, y) \vee \forall x \exists y Q(x, y)$

c) $\forall x \exists y (P(x, y) \wedge \exists z R(x, y, z))$

d) $\forall x \exists y (P(x, y) \rightarrow Q(x, y))$

31. As we push the negation symbol toward the inside, each quantifier it passes must change its type. For logical connectives we either use De Morgan's laws or recall that $\neg(p \rightarrow q) \equiv p \wedge \neg q$.

a)

$$\begin{aligned} \neg\forall x\exists y\forall z\, T(x,y,z) &\equiv \exists x\neg\exists y\forall z\, T(x,y,z) \\ &\equiv \exists x\forall y\neg\forall z\, T(x,y,z) \\ &\equiv \exists x\forall y\exists z\, \neg T(x,y,z) \end{aligned}$$

b)

$$\begin{aligned} \neg(\forall x\exists y\, P(x,y) \vee \forall x\exists y\, Q(x,y)) &\equiv \neg\forall x\exists y\, P(x,y) \wedge \neg\forall x\exists y\, Q(x,y) \\ &\equiv \exists x\neg\exists y\, P(x,y) \wedge \exists x\neg\exists y\, Q(x,y) \\ &\equiv \exists x\forall y\, \neg P(x,y) \wedge \exists x\forall y\, \neg Q(x,y) \end{aligned}$$

c)

$$\begin{aligned} \neg\forall x\exists y(P(x,y) \wedge \exists z\, R(x,y,z)) &\equiv \exists x\neg\exists y(P(x,y) \wedge \exists z\, R(x,y,z)) \\ &\equiv \exists x\forall y\, \neg(P(x,y) \wedge \exists z\, R(x,y,z)) \\ &\equiv \exists x\forall y(\neg P(x,y) \vee \neg\exists z\, R(x,y,z)) \\ &\equiv \exists x\forall y(\neg P(x,y) \vee \forall z\, \neg R(x,y,z)) \end{aligned}$$

d)

$$\begin{aligned} \neg\forall x\exists y(P(x,y) \rightarrow Q(x,y)) &\equiv \exists x\, \neg\exists y(P(x,y) \rightarrow Q(x,y)) \\ &\equiv \exists x\forall y\, \neg(P(x,y) \rightarrow Q(x,y)) \\ &\equiv \exists x\forall y(P(x,y) \wedge \neg Q(x,y)) \end{aligned}$$

33. Rewrite each of these statements so that negations appear only within predicates (that is, so that no negation is outside a quantifier or an expression involving logical connectives).

a) $\neg\forall x \forall y P(x, y)$ **b)** $\neg\forall y \exists x P(x, y)$

c) $\neg\forall y \forall x (P(x, y) \vee Q(x, y))$

d) $\neg(\exists x \exists y \neg P(x, y) \wedge \forall x \forall y Q(x, y))$

e) $\neg\forall x (\exists y \forall z P(x, y, z) \wedge \exists z \forall y P(x, y, z))$

33. We need to use the transformations shown in Table 2 of Section 1.4, replacing $\neg\forall$ by $\exists\neg$, and replacing $\neg\exists$ by $\forall\neg$. In other words, we push all the negation symbols inside the quantifiers, changing the sense of the quantifiers as we do so, because of the equivalences in Table 2 of Section 1.4. In addition, we need to use De Morgan's laws (Section 1.3) to change the negation of a conjunction to the disjunction of the negations and to change the negation of a disjunction to the conjunction of the negations. We also use the double negation law.

a) $\exists x \exists y \neg P(x, y)$ **b)** $\exists y \forall x \neg P(x, y)$

c) We can think of this in two steps. First we transform the expression into the equivalent expression $\exists y \exists x \neg(P(x, y) \vee Q(x, y))$, and then we use De Morgan's law to rewrite this as $\exists y \exists x(\neg P(x, y) \wedge \neg Q(x, y))$.

d) First we apply De Morgan's law to write this as a disjunction: $(\neg\exists x \exists y \neg P(x, y)) \vee (\neg\forall x \forall y Q(x, y))$. Then we push the negation inside the quantifiers, and note that the two negations in front of P then cancel out $(\neg\neg P(x, y) \equiv P(x, y))$. So our final answer is $(\forall x \forall y P(x, y)) \vee (\exists x \exists y \neg Q(x, y))$.

e) First we push the negation inside the outer universal quantifier, then apply De Morgan's law, and finally push it inside the inner quantifiers: $\exists x \neg(\exists y \forall z P(x, y, z) \wedge \exists z \forall y P(x, y, z)) \equiv \exists x(\neg\exists y \forall z P(x, y, z) \vee \neg\exists z \forall y P(x, y, z)) \equiv \exists x(\forall y \exists z \neg P(x, y, z) \vee \forall z \exists y \neg P(x, y, z))$.

39. Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all integers.

a) $\forall x \forall y (x^2 = y^2 \rightarrow x = y)$
b) $\forall x \exists y (y^2 = x)$
c) $\forall x \forall y (xy \geq x)$

39. a) Since the square of a number and its additive inverse are the same, we have many counterexamples, such as $x = 2$ and $y = -2$.

b) This statement is saying that every number has a square root. If x is negative (like $x = -4$), or, since we are working in the domain of the integers, x is not a perfect square (like $x = 6$), then the equation $y^2 = x$ has no solution.

c) Since negative numbers are not larger than positive numbers, we can take something like $x = 17$ and $y = -1$ for our counterexample.

45. Determine the truth value of the statement $\forall x \exists y(xy = 1)$ if the domain for the variables consists of

a) the nonzero real numbers.

b) the nonzero integers.

c) the positive real numbers.

45. This statement says that every number has a multiplicative inverse.

a) In the universe of nonzero real numbers, this is certainly true. In each case we let $y = 1/x$.

b) Integers usually don't have inverses that are integers. If we let $x = 3$, then no integer y satisfies $xy = 1$. Thus in this setting, the statement is false.

c) As in part **(a)** this is true, since $1/x$ is positive when x is positive.

47. Show that the two statements $\neg\exists x \forall y P(x, y)$ and $\forall x \exists y \neg P(x, y)$, where both quantifiers over the first variable in $P(x, y)$ have the same domain, and both quantifiers over the second variable in $P(x, y)$ have the same domain, are logically equivalent.

47. We use the equivalences explained in Table 2 of Section 1.4, twice:

$$\neg\exists x \forall y P(x, y) \equiv \forall x \neg\forall y P(x, y) \equiv \forall x \exists y \neg P(x, y)$$